CLASS 9 SCIENCE · QUICK NOTES

CHAPTER 4

Describing Motion Around Us

Position · Distance & displacement · Speed, velocity, acceleration · Graphs · Equations of motion · Circular motion

THE CHAPTER IN 30 SECONDS

Motion means **change of position with time**, measured from a **reference point**. We describe it with three ideas: **how far** (distance / displacement), **how fast** (speed / velocity) and **how quickly velocity changes** (acceleration). Graphs show the same story as pictures, and **three equations** predict the motion when acceleration is constant.

1 Position and motion

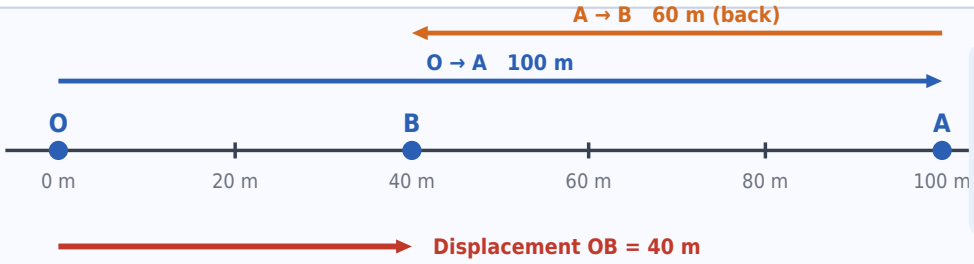
NCERT 4.1.1

- **Reference point** (origin O) = a fixed point we measure from. **Position** = distance + direction from it, at an instant.
- On a straight line: right of O is **+**, left of O is **−**.
- **In motion** = position changes with time. **At rest** = position does not change.
- **Instant of time** = one clock reading. **Time interval** = duration between two readings. They are not the same thing.

2 Distance and displacement

NCERT 4.1.2

	Distance travelled	Displacement
What is it?	Total length of the path covered	Net change in position (final − initial)
Type	Scalar: magnitude only	Vector: magnitude + direction
Can it be zero?	No, if the object has moved	Yes, e.g. when you return to the start
Compare	Always ≥ size of displacement	Size is ≤ distance
SI unit	metre (m)	metre (m)



Athlete:

Distance
= 100 + 60 = 160 m

Displacement
= OB = 40 m (+ side)

★ MEMORY HOOK

Distance = the odometer reading (all the ground you covered). **Displacement** = a straight arrow from start to end.

REMEMBER

Distance and magnitude of displacement are **equal only** for motion in a straight line without turning back. Scalars: number + unit. Vectors: number + unit + direction.

3 Speed and velocity

NCERT 4.1.3

average speed = distance ÷ time
total distance ÷ time interval · scalar (no direction)

v_{av} = displacement ÷ time
s ÷ t · vector (+ / − gives direction)

- **Uniform motion:** equal distances in equal time intervals (constant speed). **Non-uniform:** unequal distances in equal intervals.
- Velocity is the **rate of change of position**; its direction is the direction of displacement. SI unit of both: **m s^{-1}** (also km h^{-1}).
- In a straight line without turning back, average speed = size of average velocity.
- A round trip gives **average velocity = 0** but average speed ≠ 0. A speedometer shows (almost) the size of velocity at an instant.

★ MEMORY HOOK
km h⁻¹ → m s⁻¹: multiply by 5/18 (÷ 3.6).
Handy: 36 km h⁻¹ = 10 m s⁻¹, 54 = 15, 72 = 20, 108 = 30.

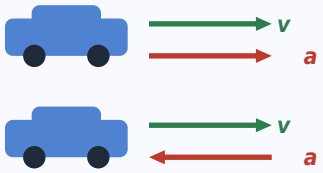
✎ Swimmer (Example 4.2)
50 m there and back, in 50 s.
Average speed = 50 ÷ 50 = **1 m s⁻¹**
Average velocity = 0 ÷ 50 = **0** (displacement is zero)

4 Acceleration

NCERT 4.1.4

a = (v − u) ÷ (t₂ − t₁)
= change in velocity ÷ time
SI unit: m s⁻² (vector)

Direction of acceleration



Speeding up
a same way as v

Slowing down
a opposite to v (negative)

- **Speeding up** → acceleration along velocity (+). **Slowing down** → acceleration opposite to velocity (−).
- Acceleration comes from a change in **speed, direction, or both**.
- **Constant velocity ⇒ a = 0**, however fast. Acceleration measures *how quickly velocity changes*, not how fast you go.
- **Free fall:** g = 9.8 m s⁻² downward. Velocity after each second: 9.8, 19.6, 29.4, 39.2 m s⁻¹ (equal gain each second).

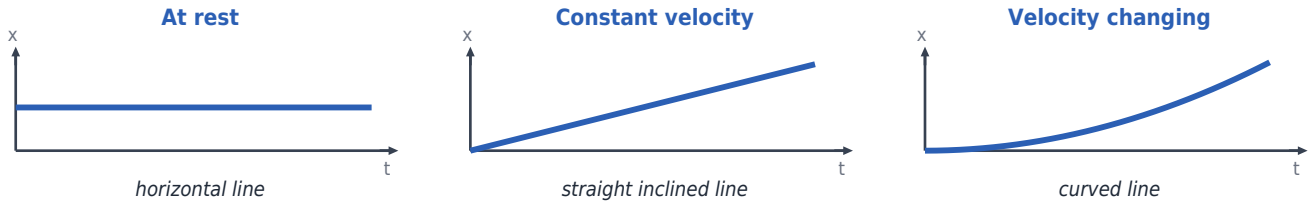
✎ Bus (Example 4.3)
Accelerating: u = 36 km h⁻¹ = 10 m s⁻¹, v = 54 km h⁻¹ = 15 m s⁻¹, t = 10 s → a = (15 − 10) ÷ 10 = **0.5 m s⁻²** (along velocity).
Braking: u = 15, v = 0, t = 5 s → a = (0 − 15) ÷ 5 = **−3 m s⁻²** (minus sign: opposite to velocity).

5 Graphs of motion

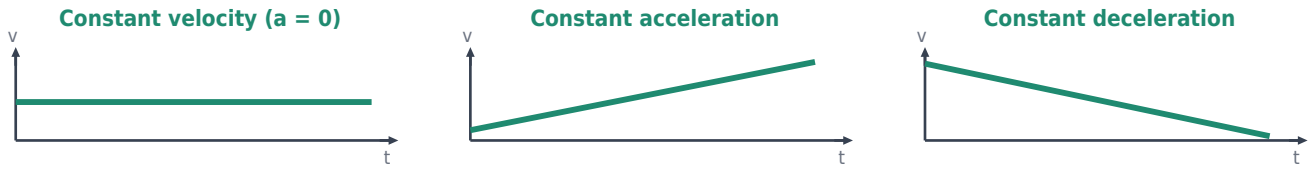
NCERT 4.2

A graph shows how one quantity changes with time. It is **not a route map**. Time goes on the x-axis. (In this chapter graphs are for straight-line motion in one direction, so distance = displacement.)

POSITION–TIME (x–t) GRAPHS · slope = velocity



VELOCITY–TIME (v–t) GRAPHS · slope = acceleration · area = displacement



x–t slope = v
v–t slope = a
v–t area = s
the three graph rules

✎ Reading graphs
x–t: (2 s, 40 m) and (4 s, 80 m) → v = 40 ÷ 2 = **20 m s⁻¹**. A steeper line = faster.
v–t slope: 5 m s⁻¹ at 10 s and 10 m s⁻¹ at 20 s → a = 5 ÷ 10 = **0.5 m s⁻²**.
v–t area: 20 m s⁻¹ for 6 s → rectangle = **120 m**. Accelerating 10 s → 20 s: rectangle 5 × 10 = 50 plus triangle ½ × 10 × 5 = 25 → **75 m**.

★ MEMORY HOOK
"Slope, slope, area": the slope of x–t is velocity, the slope of v–t is acceleration, the area under v–t is displacement.

6 Equations of motion (constant acceleration)

NCERT 4.3

v = u + at
no s in it

s = ut + ½at²
no v in it

v² = u² + 2as
no t in it

- Five quantities: **s** displacement, **u** initial velocity, **v** final velocity, **a** acceleration, **t** time ("SUVAT"). Pick the equation that **does not contain the quantity you neither know nor need**.
- **Valid only when acceleration is constant.** Choose a positive direction and keep it; the signs of u, v, a, s show direction.
- **Where they come from:** slope of the v–t line gives $v = u + at$; area (rectangle + triangle) gives $s = ut + \frac{1}{2}at^2$; removing t gives $v^2 = u^2 + 2as$.
- *Extra forms (Journey Beyond):* $s = vt - \frac{1}{2}at^2$ and $s = \frac{1}{2}(u + v)t$.

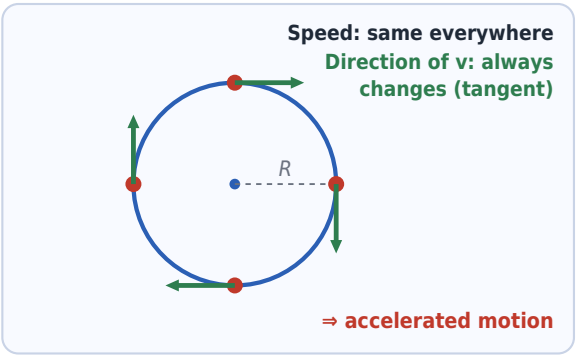
✎ **Braking distance (Example 4.8)**
$a = -4\ m\ s^{-2}$, $v = 0$. From $v^2 = u^2 + 2as$: $0 = u^2 - 8s \rightarrow$ **$s = u^2 \div 8$**.
$54\ km\ h^{-1}$ ($15\ m\ s^{-1}$) → **28.1 m**; $108\ km\ h^{-1}$ ($30\ m\ s^{-1}$) → **112.5 m**.

★ **MEMORY HOOK**
Double the speed → 4 × the stopping distance ($s \propto u^2$). Safe distance also depends on road (wet/dry), brakes and the driver's reaction time.

7 Motion in a plane: uniform circular motion

NCERT 4.4

- **Motion in a plane** (2-D): overtaking vehicle, kicked ball, satellite. Bird or aircraft in the sky = 3-D (higher grades).
- Going once round a circle of radius R: **distance = 2πR**, **displacement = 0**.
- Average speed = **2πR ÷ T** (T = time for one revolution); average velocity over T = **0**.
- **Uniform circular motion:** constant speed on a circular path.
- Speed is constant but the **direction of velocity keeps changing** (always along the tangent), so velocity changes ⇒ the motion **is accelerated**.
- Square track (4 turns) → hexagon (6 turns) → more sides → circle (turning all the time).



GOOD TO KNOW
Uniform circular motion is an **idealised model** (real speed and path are rarely perfect) but it is the base for planets circling the Sun and a vehicle taking a turn. When the ring is lifted, the marble flies off in a **straight line along the tangent**.

✎ How to solve a numerical + solved textbook problems

Revise, Reflect, Refine

5-STEP METHOD
1. Write what is given (u, v, a, t, s) with units. **2.** Convert km h^{-1} → m s^{-1} (× 5/18). **3.** Fix a positive direction. **4.** Pick the equation without the unwanted quantity. **5.** Substitute, solve, check the sign and the unit.

Problem	Working	Answer
Motorbike: $u = 28\ m\ s^{-1}$, stops after 98 m. Find a and time to stop.	$v^2 = u^2 + 2as \rightarrow 0 = 784 + 196a$; then $t = (v - u) \div a = (0 - 28) \div (-4)$	$a = -4\ m\ s^{-2}$, $t = 7\ s$
Truck slows from 54 km h^{-1} to 36 km h^{-1} in 36 s. Distance covered?	$u = 15\ m\ s^{-1}$, $v = 10\ m\ s^{-1}$; $s = \frac{1}{2}(u + v)t = \frac{1}{2} \times 25 \times 36$	$s = 450\ m$
Car: rest → 20 m s^{-1} in 5 s, then 20 m s^{-1} for 10 s, then stops in 6 s. Total distance?	Area under the v–t graph: $\frac{1}{2} \times 5 \times 20 + 20 \times 10 + \frac{1}{2} \times 6 \times 20 = 50 + 200 + 60$	$s = 310\ m$
Bus at 36 km h^{-1} sees an obstacle 30 m ahead. Reaction time 0.5 s, then $a = -2.5\ m\ s^{-2}$. Does it stop in time?	Reaction: $10 \times 0.5 = 5$ m. Braking: $s = u^2 \div 2\lvert a\rvert = 100 \div 5 = 20$ m. Total = 25 m	25 m < 30 m, so yes (5 m to spare)

★ Quick revision

one-page summary

Quantity	Formula	Unit	Scalar / vector
Distance	length of path	m	scalar
Displacement	final position − initial position	m	vector
Average speed	total distance ÷ time interval	m s^{-1}	scalar
Average velocity	displacement ÷ time interval (s ÷ t)	m s^{-1}	vector
Average acceleration	(v − u) ÷ t	m s^{-2}	vector
Equations of motion	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$		constant a only
Circular motion (1 round)	distance 2πR, displacement 0, speed 2πR ÷ T		

10 POINTS TO REMEMBER

1. Position needs a **reference point** and a **direction**; motion = position changing with time.
2. **Distance ≥ |displacement|**; they are equal only in straight-line motion without turning back.
3. Displacement, velocity, acceleration are **vectors**; distance and speed are scalars.
4. Average speed uses **distance**; average velocity uses **displacement**.
5. Acceleration is a change in velocity: **speed, direction, or both**. Constant velocity means a = 0.
6. Slowing down ⇒ acceleration is **opposite** to velocity (negative).
7. x–t: **horizontal = rest**, straight slope = constant velocity, curve = changing velocity.
8. v–t: **slope = acceleration**, **area = displacement**.
9. The three equations hold **only for constant acceleration**; convert km h^{-1} to m s^{-1} first.
10. In uniform circular motion speed is constant but velocity changes, so it is **accelerated**.

✗ COMMON MISTAKE

- Giving the path length as the displacement (or forgetting the direction of displacement).
- Saying average velocity is zero means the object did not move (a round trip: $v_{av} = 0$, speed ≠ 0).
- Thinking a fast object must be accelerating; or that a graph is the path of the object.
- Using v = u + at etc. when acceleration is not constant, or mixing km h^{-1} with m s^{-1} in one equation.

TEST YOURSELF (cover the green column)

Q1. An athlete runs 100 m forward then 60 m back along a straight track. Distance? Displacement?	**160 m ; 40 m**
Q2. A car goes 200 km north in 3 h, then 200 km south in 2 h. Average speed? Average velocity?	**80 km h^{-1} ; 0**
Q3. Convert 72 km h^{-1} into m s^{-1}.	**72 × 5/18 = 20 m s^{-1}**
Q4. A car starts from rest and reaches 24 m s^{-1} in 6 s. Find a and the distance covered.	**a = 4 m s^{-2} ; s = ½ × 4 × 6^2 = 72 m**
Q5. A bike at 20 m s^{-1} brakes with a = −4 m s^{-2}. Stopping distance?	**0 = 400 − 8s → s = 50 m**